

Homework Solution: One-Way ANOVA, Post-Hoc Tests, and Friedman Test

Emi-Marian Ionescu

Double-check summary

I checked the numerical work against the original examples and recomputed the results independently in Python. The data for the ANOVA example are the reaction-time data with three groups of size 10, with sums 145, 519, and 252. The data for the Friedman example are the 21×4 objective-function table from the Villegas metaheuristics example. The calculations below are consistent with those sources and with the Python output.

The ANOVA part uses Tukey's HSD as the main post-hoc method. I also include a Scheffe check, because the homework statement mentions Tukey or Scheffe. Both methods give the same substantive conclusion: all three group means are significantly different.

1 One-Way ANOVA and post-hoc analysis

1.1 Data and hypotheses

The data are reaction-time scores, measured in seconds, for three independent groups.

Drug A	Drug B	Placebo
12	45	25
15	54	26
16	39	28
13	65	22
14	34	26
15	63	27
17	55	23
11	51	26
18	53	24
14	60	25

The null and alternative hypotheses are

$$H_0 : \mu_1 = \mu_2 = \mu_3, \quad H_1 : \text{at least one group mean is different.}$$

We use significance level $\alpha = 0.05$ for the ANOVA and post-hoc tests.

1.2 Manual ANOVA calculations

The group sizes are equal:

$$n_1 = n_2 = n_3 = 10, \quad k = 3, \quad N = 30.$$

The group means are

$$\bar{x}_1 = 14.5, \quad \bar{x}_2 = 51.9, \quad \bar{x}_3 = 25.2.$$

The grand mean is

$$\bar{x} = \frac{145 + 519 + 252}{30} = \frac{916}{30} = 30.5333.$$

The between-groups sum of squares is

$$SS_B = \sum_{j=1}^k n_j (\bar{x}_j - \bar{x})^2 = 7420.4667.$$

The within-groups sum of squares is

$$SS_W = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_j)^2 = 983.0000.$$

The total sum of squares is

$$SS_T = SS_B + SS_W = 8403.4667.$$

The degrees of freedom are

$$df_B = k - 1 = 2, \quad df_W = N - k = 27, \quad df_T = N - 1 = 29.$$

Therefore,

$$MS_B = \frac{SS_B}{df_B} = \frac{7420.4667}{2} = 3710.2333,$$

$$MS_W = \frac{SS_W}{df_W} = \frac{983.0000}{27} = 36.4074.$$

The ANOVA statistic is

$$F = \frac{MS_B}{MS_W} = \frac{3710.2333}{36.4074} = 101.9087.$$

Source	SS	df	MS	F	p -value
Between groups	7420.4667	2	3710.2333	101.9087	$2.626 \cdot 10^{-13}$
Within groups	983.0000	27	36.4074	—	—
Total	8403.4667	29	—	—	—

The critical value is

$$F_{0.95;2,27} = 3.3541.$$

Since

$$101.9087 > 3.3541$$

and also

$$p = 2.626 \cdot 10^{-13} < 0.05,$$

we reject H_0 . The group means are not all equal.

1.3 Tukey HSD post-hoc test

Because the ANOVA is significant, we perform pairwise post-hoc comparisons. Since the groups have equal size $n = 10$, Tukey's minimum significant difference is

$$\text{HSD} = q_{0.95;k,df_W} \sqrt{\frac{MS_W}{n}}.$$

Here,

$$q_{0.95;3,27} = 3.5064,$$

so

$$\text{HSD} = 3.5064 \sqrt{\frac{36.4074}{10}} = 6.6905.$$

Therefore, a pair of means is significantly different if its absolute difference is larger than 6.6905.

Comparison	Mean difference	Tukey adjusted p -value	Significant?
Drug A vs Drug B	37.4	$2.55 \cdot 10^{-13}$	Yes
Drug A vs Placebo	10.7	0.0014	Yes
Drug B vs Placebo	26.7	$5.29 \cdot 10^{-10}$	Yes

All pairwise differences are larger than the Tukey HSD threshold. Therefore, all three pairs are significantly different at $\alpha = 0.05$.

1.4 Scheffe check

The homework allows Tukey or Scheffe. As a check, the Scheffe critical difference for pairwise comparisons is

$$CD_S = \sqrt{(k-1)F_{0.95;k-1,N-k}MS_W \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}.$$

Because all group sizes are 10,

$$CD_S = \sqrt{2 \cdot 3.3541 \cdot 36.4074 \cdot \left(\frac{1}{10} + \frac{1}{10} \right)} = 6.9890.$$

Comparison	Absolute mean difference	Scheffe critical difference	Significant?
Drug A vs Drug B	37.4	6.9890	Yes
Drug A vs Placebo	10.7	6.9890	Yes
Drug B vs Placebo	26.7	6.9890	Yes

Thus, Scheffe's method confirms the same conclusion as Tukey's method.

1.5 Conclusion for the ANOVA problem

At $\alpha = 0.05$, we reject the null hypothesis that all three population means are equal. The post-hoc tests show that all three groups differ significantly from each other. The observed ordering of sample means is

$$\bar{x}_{\text{Drug B}} = 51.9 > \bar{x}_{\text{Placebo}} = 25.2 > \bar{x}_{\text{Drug A}} = 14.5.$$

2 Friedman test for the metaheuristics example

2.1 Data and hypotheses

The data consist of objective-function values for $k = 4$ metaheuristics over $b = 21$ problem instances. Lower objective-function values are better. Therefore, inside each row, rank 1 is assigned to the smallest value and rank 4 to the largest value.

The hypotheses are

H_0 : each ranking of the metaheuristics is equally likely,

H_1 : at least one metaheuristic tends to perform differently.

For this part we follow the paper and use significance level $\alpha = 0.01$.

Instance	X_{b1}	R_{b1}	X_{b2}	R_{b2}	X_{b3}	R_{b3}	X_{b4}	R_{b4}
1	565.02	1	567.98	3	568.86	4	565.99	2
2	662.84	4	619.35	3	617.48	2	614.23	1
3	664.73	4	629.59	3	620.50	2	618.04	1
4	857.84	4	809.13	2	817.71	3	803.51	1
5	949.98	4	858.98	3	858.95	2	841.63	1
6	1084.82	4	949.89	2	942.60	1	961.47	3
7	837.80	3	832.91	2	838.50	4	830.48	1
8	906.16	4	881.26	2	882.70	3	876.21	1
9	1000.27	4	955.95	3	921.97	2	918.45	1
10	1076.88	4	1052.65	2	1074.38	3	1050.11	1
11	1170.17	4	1107.47	2	1108.88	3	1100.95	1
12	1217.01	4	1184.58	3	1166.59	2	1158.88	1
13	1364.50	4	1296.33	1	1340.98	3	1305.83	2
14	1464.20	4	1384.13	3	1367.91	2	1354.04	1
15	1544.21	4	1488.71	3	1454.91	2	1437.52	1
16	1064.89	4	1003.00	1	1007.26	3	1003.07	2
17	1104.67	4	1042.79	3	1035.23	1	1042.61	2
18	1202.00	4	1141.94	3	1110.13	1	1118.63	2
19	887.22	4	813.98	1	823.01	3	819.81	2
20	963.06	4	852.89	1	859.06	2	860.12	3
21	952.29	4	914.04	2	915.38	3	909.06	1

2.2 Rank summaries

The rank sums, average ranks, and sums of squared ranks are:

Metaheuristic	Rank sum R_j	Average rank	Sum of squared ranks
M_1	80	3.8095	314
M_2	48	2.2857	122
M_3	51	2.4286	139
M_4	31	1.4762	55

The total sum of squared ranks is

$$A_2 = 314 + 122 + 139 + 55 = 630.$$

Also,

$$B_2 = \frac{1}{b} \sum_{j=1}^k R_j^2 = \frac{80^2 + 48^2 + 51^2 + 31^2}{21} = 584.0952.$$

2.3 Friedman test statistic following the paper

The paper uses the following F -approximation statistic:

$$T_2 = \frac{(b-1) \left(B_2 - \frac{bk(k+1)^2}{4} \right)}{A_2 - B_2}.$$

Substituting $b = 21$, $k = 4$, $A_2 = 630$, and $B_2 = 584.0952$ gives

$$T_2 = \frac{20 \left(584.0952 - \frac{21 \cdot 4 \cdot 25}{4} \right)}{630 - 584.0952} = 25.7469.$$

The critical value is

$$F_{0.99;3,60} = 4.1259.$$

Since

$$25.7469 > 4.1259,$$

we reject H_0 at $\alpha = 0.01$.

As an additional Python check, `scipy.stats.friedmanchisquare` gives the chi-square version of the Friedman test:

$$\chi^2 = 35.4571, \quad p = 9.753 \cdot 10^{-8}.$$

This also rejects H_0 .

2.4 Post-hoc paired comparisons after Friedman

The paper compares absolute rank-sum differences. A pair of metaheuristics is declared significantly different if

$$|R_i - R_j| > t_{1-\alpha/2, (b-1)(k-1)} \sqrt{\frac{2b(A_2 - B_2)}{(b-1)(k-1)}}.$$

Here,

$$(b-1)(k-1) = 20 \cdot 3 = 60,$$

$$t_{0.995,60} = 2.6603,$$

so the critical difference is

$$2.6603 \sqrt{\frac{2 \cdot 21(630 - 584.0952)}{20 \cdot 3}} = 15.0802.$$

Comparison	$ R_i - R_j $	Different?
M_1 vs M_2	32	Yes
M_1 vs M_3	29	Yes
M_1 vs M_4	49	Yes
M_2 vs M_3	3	No
M_2 vs M_4	17	Yes
M_3 vs M_4	20	Yes

2.5 Conclusion for the Friedman problem

Since lower ranks are better, the best average rank is obtained by M_4 :

$$\bar{R}_{M_4} = 1.4762.$$

The post-hoc comparisons show that M_1 is significantly worse than all the other metaheuristics, M_2 and M_3 are not significantly different, and M_4 is significantly better than each of the other metaheuristics.

References

- W. Montelpare et al., *Applied Statistics in Healthcare Research*, Chapter 31, “The One Way Analysis of Variance and Post Hoc Tests.” URL: <https://pressbooks.library.upei.ca/montelpare/chapter/the-analysis-of-variance-and-post-hoc-tests/>
- J. G. Villegas, “Using Nonparametric Test to Compare the Performance of Metaheuristics.” URL: <https://juangvillegas.com/wp-content/uploads/2011/08/friedman-test-24062011.pdf>
- SciPy documentation, `scipy.stats.f_oneway` and `scipy.stats.friedmanchisquare`.
- statsmodels documentation, `statsmodels.stats.multicomp.pairwise_tukeyhsd`.

A Python code used to double-check the calculations

```
"""
Homework solution: One-Way ANOVA + Tukey/Scheffe post-hoc tests,
and Friedman test + post-hoc rank-sum comparisons.

Required packages:
    pip install numpy scipy pandas statsmodels
"""

import itertools
import numpy as np
import pandas as pd
from scipy import stats
from scipy.stats import studentized_range
from statsmodels.stats.multicomp import pairwise_tukeyhsd

ALPHA_ANOVA = 0.05
ALPHA_FRIEDMAN = 0.01

def anova_tukey_scheffe():
    # Reaction time scores from the Montelpare chapter example.
    drug_a = np.array([12, 15, 16, 13, 14, 15, 17, 11, 18, 14], dtype=float)
    drug_b = np.array([45, 54, 39, 65, 34, 63, 55, 51, 53, 60], dtype=float)
    placebo = np.array([25, 26, 28, 22, 26, 27, 23, 26, 24, 25], dtype=float)

    groups = [drug_a, drug_b, placebo]
    names = ["Drug A", "Drug B", "Placebo"]

    all_values = np.concatenate(groups)
    labels = np.repeat(names, [len(g) for g in groups])
```

```

k = len(groups)
N = len(all_values)
grand_mean = all_values.mean()
means = np.array([g.mean() for g in groups])

ss_between = sum(len(g) * (g.mean() - grand_mean) ** 2 for g in groups)
ss_within = sum(((g - g.mean()) ** 2).sum() for g in groups)
ss_total = ((all_values - grand_mean) ** 2).sum()

df_between = k - 1
df_within = N - k
df_total = N - 1
ms_between = ss_between / df_between
ms_within = ss_within / df_within
F_manual = ms_between / ms_within
F_scipy, p_scipy = stats.f_oneway(*groups)
F_crit = stats.f.ppf(1 - ALPHA_ANOVA, df_between, df_within)

print("=== 1. ONE-WAY ANOVA ===")
print(pd.DataFrame({"Group": names, "n": [len(g) for g in groups], "Mean": means}))
print(f"Grand mean = {grand_mean:.6f}\n")

anova_table = pd.DataFrame({
    "Source": ["Between groups", "Within groups", "Total"],
    "SS": [ss_between, ss_within, ss_total],
    "df": [df_between, df_within, df_total],
    "MS": [ms_between, ms_within, np.nan],
    "F": [F_manual, np.nan, np.nan],
    "p-value": [p_scipy, np.nan, np.nan],
})
print(anova_table.to_string(index=False))
print(f"\nSciPy f_oneway: F = {F_scipy:.6f}, p = {p_scipy:.6e}")
print(f"F critical at alpha = 0.05: {F_crit:.6f}")
print("Decision: reject H0 at alpha = 0.05.\n")

print("=== Tukey HSD post-hoc test ===")
tukey = pairwise_tukeyhsd(all_values, labels, alpha=ALPHA_ANOVA)
print(tukey)

qcrit = studentized_range.ppf(1 - ALPHA_ANOVA, k, df_within)
tukey_hsd = qcrit * np.sqrt(ms_within / len(drug_a))
print(f"\nTukey q critical = {qcrit:.6f}")
print(f"Tukey minimum significant difference = {tukey_hsd:.6f}")

print("\n=== Scheffe post-hoc check ===")
scheffe_cd = np.sqrt((k - 1) * F_crit * ms_within * (1 / len(drug_a) + 1 / len(drug_b)))
print(f"Scheffe critical difference for equal n = {scheffe_cd:.6f}")
for (i, (name_i, group_i)), (j, (name_j, group_j)) in itertools.combinations(enumerate(zip(names, groups)),
    2):
    diff = abs(group_i.mean() - group_j.mean())
    F_pair = diff ** 2 / (ms_within * (1 / len(group_i) + 1 / len(group_j))) / (k - 1)
    p_pair = stats.f.sf(F_pair, k - 1, df_within)
    print(f"{name_i} vs {name_j}: difference = {diff:.6f}, "
        f"Scheffe F = {F_pair:.6f}, p = {p_pair:.6e}, "
        f"significant = {diff > scheffe_cd}")

def friedman_metaheuristics():
    # Objective values for 4 metaheuristics over 21 instances.
    # Lower values are better, so rank 1 is assigned to the smallest value in each row.
    X = np.array([
        [565.02, 567.98, 568.86, 565.99],
        [662.84, 619.35, 617.48, 614.23],
        [664.73, 629.59, 620.50, 618.04],
        [857.84, 809.13, 817.71, 803.51],
        [949.98, 858.98, 858.95, 841.63],
        [1084.82, 949.89, 942.60, 961.47],
        [837.80, 832.91, 838.50, 830.48],
        [906.16, 881.26, 882.70, 876.21],
        [1000.27, 955.95, 921.97, 918.45],
        [1076.88, 1052.65, 1074.38, 1050.11],
        [1170.17, 1107.47, 1108.88, 1100.95],
        [1217.01, 1184.58, 1166.59, 1158.88],
        [1364.50, 1296.33, 1340.98, 1305.83],
    ])

```

```

[1464.20, 1384.13, 1367.91, 1354.04],
[1544.21, 1488.71, 1454.91, 1437.52],
[1064.89, 1003.00, 1007.26, 1003.07],
[1104.67, 1042.79, 1035.23, 1042.61],
[1202.00, 1141.94, 1110.13, 1118.63],
[887.22, 813.98, 823.01, 819.81],
[963.06, 852.89, 859.06, 860.12],
[952.29, 914.04, 915.38, 909.06],
], dtype=float)

b, k = X.shape
names = [f"M{j}" for j in range(1, k + 1)]
ranks = np.array([stats.rankdata(row, method="average") for row in X])

rank_sums = ranks.sum(axis=0)
avg_ranks = ranks.mean(axis=0)
squared_rank_sums = (ranks ** 2).sum(axis=0)

A2 = (ranks ** 2).sum()
B2 = (rank_sums ** 2).sum() / b
T2 = ((b - 1) * (B2 - b * k * (k + 1) ** 2 / 4)) / (A2 - B2)
Fcrit = stats.f.ppf(1 - ALPHA_FRIEDMAN, k - 1, (b - 1) * (k - 1))

chi2_stat, chi2_p = stats.friedmanchisquare(*[X[:, j] for j in range(k)])

print("\n\n=== 2. FRIEDMAN TEST ===")
print(pd.DataFrame({
    "Metaheuristic": names,
    "Rank sum": rank_sums,
    "Average rank": avg_ranks,
    "Sum of squared ranks": squared_rank_sums,
}).to_string(index=False))

print(f"\nA2 = {A2:.6f}")
print(f"B2 = {B2:.6f}")
print(f"Paper F-approximation T2 = {T2:.6f}")
print(f"F critical at alpha = 0.01 with df = (3, 60): {Fcrit:.6f}")
print("Decision by paper method: reject H0.\n")

print(f"SciPy friedmanchisquare: chi-square = {chi2_stat:.6f}, p = {chi2_p:.6e}")
print("Decision by SciPy method: reject H0.\n")

df = (b - 1) * (k - 1)
tcrit = stats.t.ppf(1 - ALPHA_FRIEDMAN / 2, df)
critical_difference = tcrit * np.sqrt((2 * b * (A2 - B2)) / ((b - 1) * (k - 1)))
print("=== Post-hoc paired comparisons after Friedman ===")
print(f"t critical = {tcrit:.6f}")
print(f"Critical difference for rank sums = {critical_difference:.6f}")

rows = []
for i in range(k):
    for j in range(i + 1, k):
        diff = abs(rank_sums[i] - rank_sums[j])
        rows.append({
            "Comparison": f"{names[i]} vs {names[j]}",
            "|Rank sum difference|": diff,
            "Different?": diff > critical_difference,
        })
print(pd.DataFrame(rows).to_string(index=False))
print("\nConclusion: M4 is best; M1 is worse than all the others; M2 and M3 are not significantly different
.")

if __name__ == "__main__":
    anova_tukey_scheffe()
    friedman_metaheuristics()

```